

08_control_flow_statements

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1 *Control Flow Statement*

2 If elif else

```
[1]: age = 17
if age >= 18:
    print("You are eligible to vote.")
elif age <= 5:
    print("you are too young to vote!")
elif age == 17:
    print("You are almost there! Just one more year to go.")
else:
    print("Abhi chotay ho ap.")
```

You are almost there! Just one more year to go.

```
[2]: age = 19
if age >= 18:
    print("You are eligible to vote.")
elif age <= 5:
    print("you are too young to vote!")
elif age == 17:
    print("You are almost there! Just one more year to go.")
else:
    print("Abhi chotay ho ap.")
```

You are eligible to vote.

```
[3]: age = 5
if age >= 18:
    print("You are eligible to vote.")
elif age <= 5:
    print("you are too young to vote!")
elif age == 17:
    print("You are almost there! Just one more year to go.")
else:
    print("Abhi chotay ho ap.")
```

you are too young to vote!

```
[4]: age = 9
    if age >= 18:
        print("You are eligible to vote.")
    elif age <= 5:
        print("you are too young to vote!")
    elif age == 17:
        print("You are almost there! Just one more year to go.")
    else:
        print("Abhi chotay ho ap.")
```

Abhi chotay ho ap.

```
[5]: score = 85
    if score >= 90:
        print("Excellent! You got an A.")
    elif score >= 80:
        print("Grade B ")
    elif score >= 70:
        print("Grade C")
    else:
        print("Fail or Grade F")
```

Grade B

```
[6]: score = 95
    if score >= 90:
        print("Excellent! You got an A.")
    elif score >= 80:
        print("Grade B ")
    elif score >= 70:
        print("Grade C")
    else:
        print("Fail or Grade F")
```

Excellent! You got an A.

```
[7]: score = 75
    if score >= 90:
        print("Excellent! You got an A.")
    elif score >= 80:
        print("Grade B ")
    elif score >= 70:
        print("Grade C")
    else:
        print("Fail or Grade F")
```

Grade C

```
[8]: score = 60
    if score >= 90:
        print("Excellent! You got an A.")
    elif score >= 80:
        print("Grade B ")
    elif score >= 70:
        print("Grade C")
    else:
        print("Grade F")
```

Grade F

3 if elif else with relational operators

```
[9]: 18 == 16+2
    18 != 16+5
    18 > 16+1
    18 < 16+3
```

[9]: True

3.0.1 1. Sign of equality is “==”

3.0.2 2. Sign of inequality is “!=”

3.0.3 3. Sign of greater than is “>”

3.0.4 4. Sign of less than is “<”

3.0.5 5. Sign of greater than or equal to is “>=”

3.0.6 6. Sign of less than or equal to is “<=”

4 *Logical Operators in if else conditions*

4.1 Use of AND Operation with if else condition

```
[10]: age = 25
    citizen = True
    if age >= 18 and citizen:
        print("You are eligible to vote.")
    else:
        print("You are not eligible to vote.")
```

You are eligible to vote.

```
[11]: age = 17
    citizen = True
    if age >= 18 and citizen:
        print("You are eligible to vote.")
```

```
else:
    print("You are not eligible to vote.")
```

You are not eligible to vote.

4.2 Use of OR Operator with if else condition

```
[12]: day = "Sunday"
if day == "Saturday" or day == "Sunday":
    print("It's the weekend! Time to relax.")
else:
    print("It's a weekday. Time to work!")
```

It's the weekend! Time to relax.

```
[13]: day = "Tuesday"
if day == "Saturday" or day == "Sunday":
    print("It's the weekend! Time to relax.")
else:
    print("It's a weekday. Time to work!")
```

It's a weekday. Time to work!

4.3 Use of NOT Operator with if else condition

```
[14]: is_raining = False
if not is_raining:
    print("It's a nice day! You can go outside.")
else:
    print("It's raining. Better stay indoors.")
```

It's a nice day! You can go outside.

```
[15]: is_raining = True
if not is_raining:
    print("It's a nice day! You can go outside.")
else:
    print("It's raining. Better stay indoors.")
```

It's raining. Better stay indoors.

```
[16]: age = 18
license = True
if age >= 18:
    if license:
        print("You can drive.")
    else:
        print("You need a license to drive.")
else:
```

```
print("You are too young to drive.")
```

You can drive.

5 Loops

5.1 for loop

```
[17]: dishes = ["Biryani", "Pulao", "Korma", "Nihari"]
      for dish in dishes:
          print(dish)
```

Biryani
Pulao
Korma
Nihari

```
[18]: list = ["Ahmad", "Ali", "Sara", "Zainab"]
      for name in list:
          print(name)
```

Ahmad
Ali
Sara
Zainab

```
[19]: list = ["Abdullah", "Jafar", "Tahir", "Bilal", "Hamza"]
      for name in list:
          print(list)
```

['Abdullah', 'Jafar', 'Tahir', 'Bilal', 'Hamza']
['Abdullah', 'Jafar', 'Tahir', 'Bilal', 'Hamza']
['Abdullah', 'Jafar', 'Tahir', 'Bilal', 'Hamza']
['Abdullah', 'Jafar', 'Tahir', 'Bilal', 'Hamza']
['Abdullah', 'Jafar', 'Tahir', 'Bilal', 'Hamza']

```
[20]: list = ["Abdullah", "Jafar", "Tahir", "Bilal", "Hamza"]
      for name in list:
          print(name)
```

Abdullah
Jafar
Tahir
Bilal
Hamza

```
[21]: dishes = ["Biryani", "Pulao", "Korma", "Nihari"]
      for dish in dishes:
          print(f"{dish} is available in dhaba.")
```

Biryani is available in dhaba.
Pulao is available in dhaba.
Korma is available in dhaba.
Nihari is available in dhaba.

5.2 while loop

```
[22]: cups_of_chai = 0
      while cups_of_chai < 3:
          cups_of_chai += 1
          print(f"chai cup {cups_of_chai} is served.")
      print("I am full now, no more chai for me!")
```

chai cup 1 is served.
chai cup 2 is served.
chai cup 3 is served.
I am full now, no more chai for me!

```
[23]: cold_drinks = 0
      while cold_drinks < 5:
          cold_drinks += 1
          print("Cold drink is served.")
      print("I am full now, no more cold drinks for me!")
```

Cold drink is served.
Cold drink is served.
Cold drink is served.
Cold drink is served.
Cold drink is served.
I am full now, no more cold drinks for me!

```
[24]: cold_drinks = 0
      while cold_drinks < 5:
          cold_drinks += 1
          print("Cold drink is served.")
      print("I am full now, no more cold drinks for me!")
```

Cold drink is served.
Cold drink is served.
Cold drink is served.
Cold drink is served.
Cold drink is served.
I am full now, no more cold drinks for me!

```
[25]: name = "Abdul Rahman"
      while name != "Abdul Rahman":
          print(name)
      print("Welcome, Abdul Rahman!")
```

Welcome, Abdul Rahman!

5.3 Break and Continue Statement

```
[26]: cups_of_chai = 0

while cups_of_chai < 5:
    cups_of_chai += 1
    print("I want more chai.")
    if cups_of_chai == 3:
        print("Too much chai, stopping now.")
        break
    print("I have had", cups_of_chai, "cups of chai.")
```

I want more chai.
I have had 1 cups of chai.
I want more chai.
I have had 2 cups of chai.
I want more chai.
Too much chai, stopping now.

```
[27]: cups_of_chai = 0

while cups_of_chai < 5:
    cups_of_chai += 1
    print("I want more chai.")
    if cups_of_chai == 3:
        print("Too much chai, stopping now.")
        break
    print("I have had", cups_of_chai, "cups of chai.")
```

I want more chai.
I have had 1 cups of chai.
I want more chai.
I have had 2 cups of chai.
I want more chai.
Too much chai, stopping now.

5.4 continue statement

```
[28]: cups_of_chai = 0

while cups_of_chai < 5:
    cups_of_chai += 1
    print("I want more chai.")
    if cups_of_chai == 2:
        print("Too much chai, stopping now.")
        continue
    print("I have had", cups_of_chai, "cups of chai.")
```

I want more chai.

I have had 1 cups of chai.
I want more chai.
Too much chai, stopping now.
I want more chai.
I have had 3 cups of chai.
I want more chai.
I have had 4 cups of chai.
I want more chai.
I have had 5 cups of chai.

5.5 Range function in loops

```
[29]: for cup in range(1, 20, 2):  
       print("Chai is ready, Chai cup", cup, "is served.")
```

Chai is ready, Chai cup 1 is served.
Chai is ready, Chai cup 3 is served.
Chai is ready, Chai cup 5 is served.
Chai is ready, Chai cup 7 is served.
Chai is ready, Chai cup 9 is served.
Chai is ready, Chai cup 11 is served.
Chai is ready, Chai cup 13 is served.
Chai is ready, Chai cup 15 is served.
Chai is ready, Chai cup 17 is served.
Chai is ready, Chai cup 19 is served.

6 Infinite Loop in Python

```
[30]: while True:  
       print("This loop run forever.")  
       break # You need to be careful while running an infinite loop.
```

This loop run forever.

```
[31]: count = 0  
       while True:  
           count += 1  
           print("This loop run forever.")  
           if count == 5:  
               break
```

This loop run forever.
This loop run forever.
This loop run forever.
This loop run forever.
This loop run forever.

7 Pass Statement

```
[32]: cups_of_chai = 0
      while cups_of_chai < 5:
          cups_of_chai += 1
          print("I want more chai.")
          if cups_of_chai == 3:
              print("Too much chai, stopping now.")
              continue # You can use pass to do nothing and continue with the next
              ↪ iteration of the loop.
          print("I have had", cups_of_chai, "cups of chai.")
```

```
I want more chai.
I have had 1 cups of chai.
I want more chai.
I have had 2 cups of chai.
I want more chai.
Too much chai, stopping now.
I want more chai.
I have had 4 cups of chai.
I want more chai.
I have had 5 cups of chai.
```

```
[33]: cups_of_chai = 0
      while cups_of_chai < 5:
          cups_of_chai += 1
          print("I want more chai.")
          if cups_of_chai == 3:
              print("Too much chai, stopping now.")
              pass
          print("I have had", cups_of_chai, "cups of chai.")
```

```
I want more chai.
I have had 1 cups of chai.
I want more chai.
I have had 2 cups of chai.
I want more chai.
Too much chai, stopping now.
I have had 3 cups of chai.
I want more chai.
I have had 4 cups of chai.
I want more chai.
I have had 5 cups of chai.
```

7.1 try, except and finally

```
[34]: try:
      result = 10 / 0
except ZeroDivisionError:
    print("You cannot divide by zero!")
finally:
    print("This will always be executed.")
```

You cannot divide by zero!
This will always be executed.

7.1.1 Real World Example

try → paisay nikalnay ki koshish except → agar card block / balance kam ho finally → card wapas nikalna (HAR HAAL MEIN)

```
[42]: try:
      File = open("report.txt", "r")
      Data = File.read()
except FileNotFoundError:
    print("File nhi mili.")
finally:
    print("File handling is done.")
```

File nhi mili.
File handling is done.

```
[43]: try:
      print("Hello, World!")
      print(10 / 0) # This will raise an exception
except ZeroDivisionError:
    print("You cannot divide by zero!")
finally:
    print("This will always be executed.")
```

Hello, World!
You cannot divide by zero!
This will always be executed.

```
[46]: try:
      print("Hello, World!")
      print(10 / 3)
except ZeroDivisionError:
    print("You cannot divide by zero!")
finally:
    print("This will always be executed.")
```

Hello, World!
3.3333333333333335

This will always be executed.

```
[48]: try:
        print("Hello, World!")
        print(6 / 0)
    except ZeroDivisionError:
        print("Divide by zero error.")
    finally:
        print("This will always be executed.")
```

Hello, World!

Divide by zero error.

This will always be executed.

```
[ ]:
```